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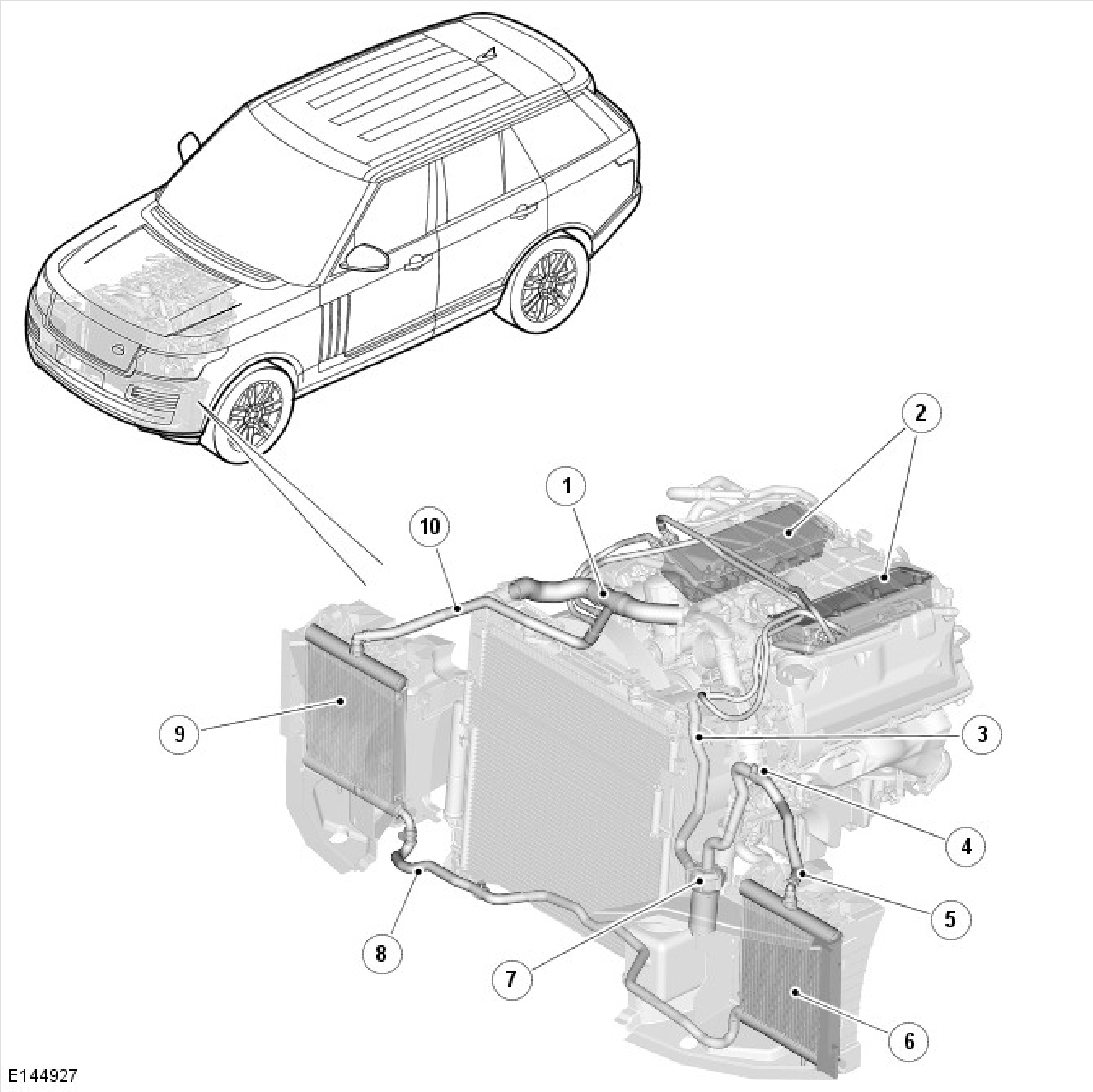
2014.0 RANGE ROVER (LG), 303-03

SUPERCHARGER COOLING - V8 S/C 5.0L PETROL

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Radiator upper hose
2	Charge air coolers
3	Hose - coolant feed to charge air coolers
4	Bleed screw
5	Hose - charge air radiator to coolant pump
6	Left charge air radiator

7	Charge air coolant pump
8	Hose - charge air radiators
9	Right charge air radiator
10	Hose - return from charge air coolers

OVERVIEW

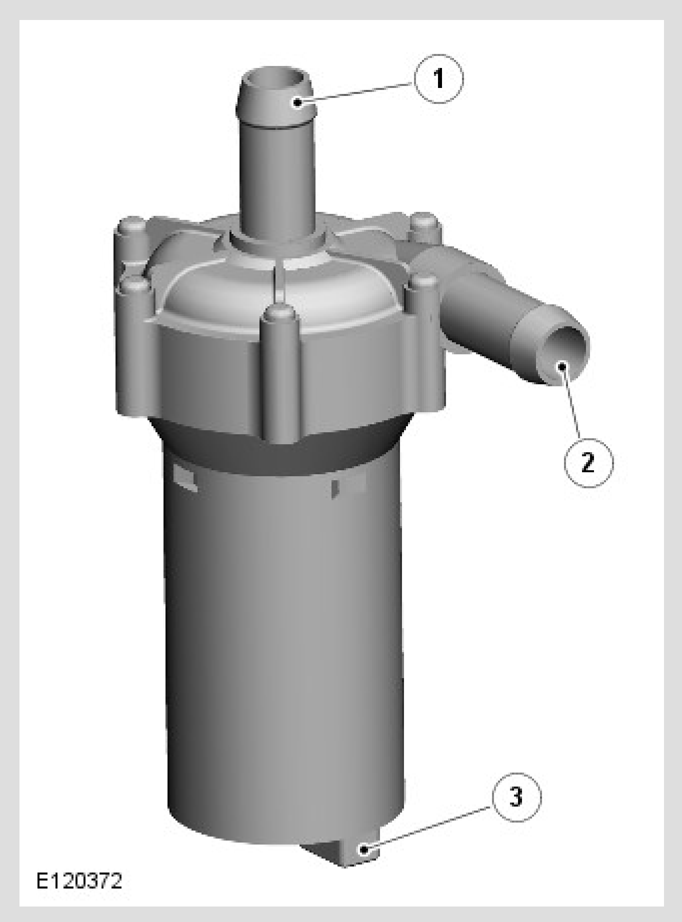
The supercharger cooling system uses engine coolant to cool the pressurized charge air from the supercharger. The supercharger cooling system consists of:

- A charge air coolant pump.
- Two charge air radiators.
- Two charge air coolers.
- Connecting hoses and pipes.

The supercharger cooling system is operationally independent of the engine cooling system, but connected to it at the radiator upper hose. The connection with the engine cooling system accommodates thermal expansion and retraction of the coolant in the supercharger cooling system, and enables filling and draining of the supercharger cooling system.

DESCRIPTION

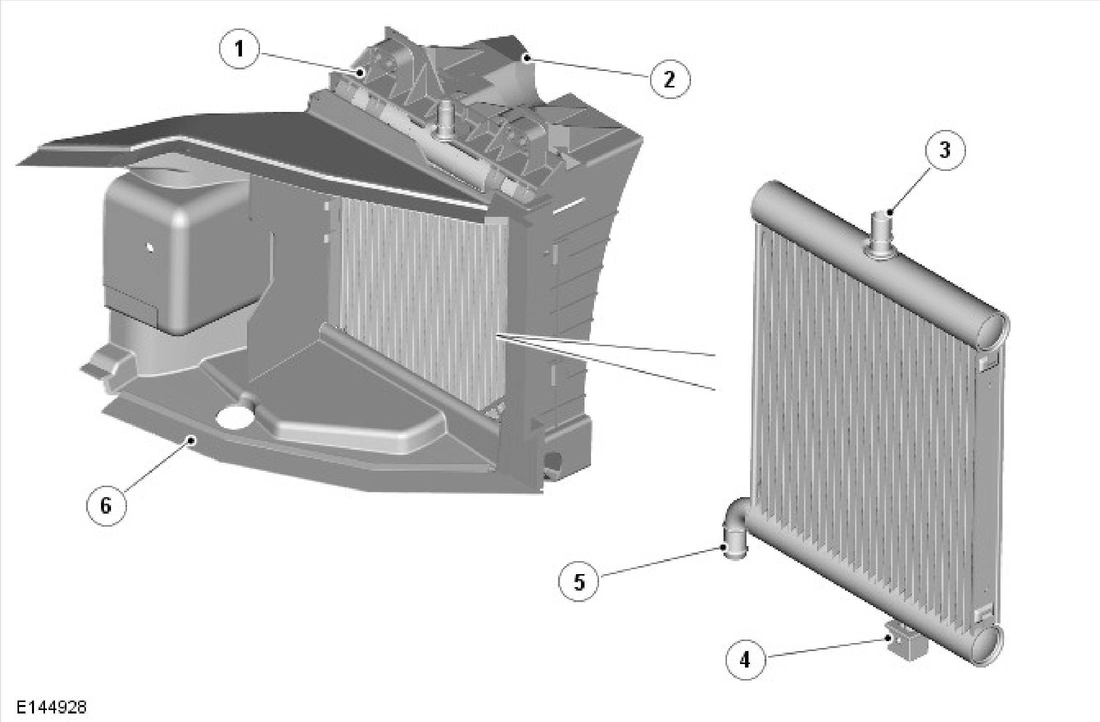
CHARGE AIR COOLANT PUMP



ITEM	DESCRIPTION
1	Coolant intake connection
2	Coolant outlet connection
3	Electrical connector

The charge air coolant pump is an electric pump attached to the rear left side of the engine cooling fan shroud. Hoses connect the intake of the charge air coolant pump to the left charge air radiator, and the outlet to the charge air coolers. An electrical connector provides the interface between the motor of the charge air coolant pump and the vehicle wiring.

CHARGE AIR RADIATORS



ITEM	DESCRIPTION
1	Radiator support bracket
2	Rear air duct
3	Coolant hose connection - outlet on left radiator, inlet on right radiator
4	Isolator
5	Coolant hose connection - inlet on left radiator, outlet on right radiator
6	Front air duct

A charge air radiator is located each side of the cooling pack, behind the outer air intakes in the front bumper. The charge air radiators are installed in air ducts attached to the front bumper armature and the front end carrier. The lower end tank of each charge air radiator is located in the air ducts by a spigot fitted with an isolator bush. The upper end tank of each charge air radiator is held in a support bracket in the air duct.

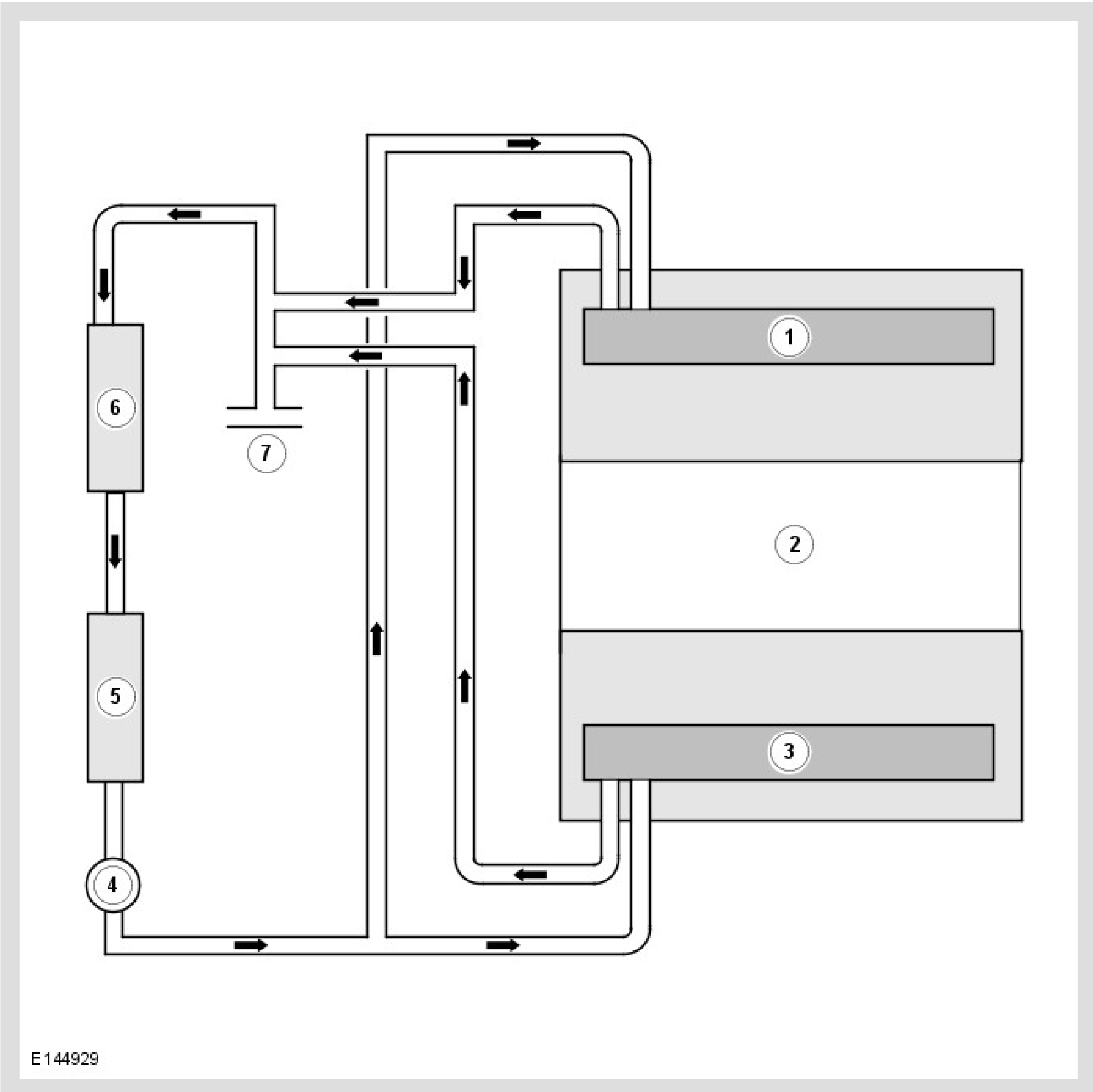
Hoses connect the two charge air radiators in series, the intake of the right radiator to the charge air coolers, and the outlet of the left radiator to the charge air coolant pump.

CHARGE AIR COOLERS

A charge air cooler is installed in each intake manifold.
For additional information, refer to: Intake Air Distribution and Filtering (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Description and Operation).

OPERATION

Supercharger Cooling Flow Diagram



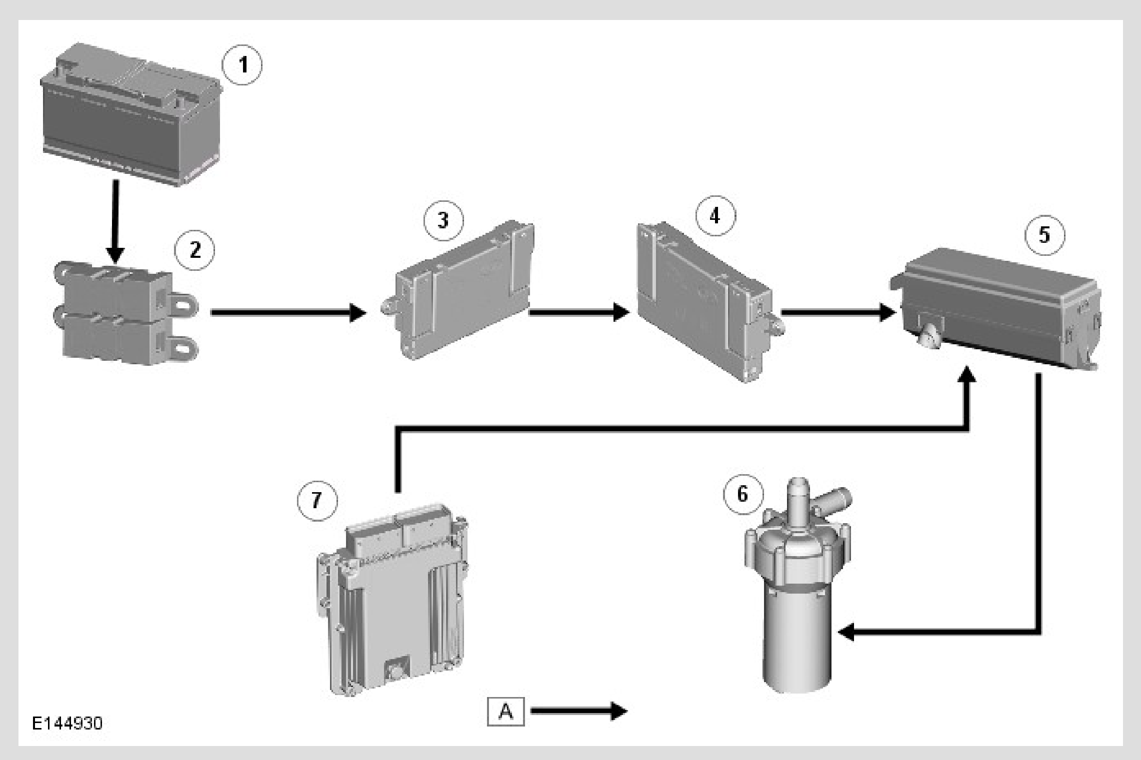
ITEM	DESCRIPTION

1	Right charge air cooler
2	Engine
3	Left charge air cooler
4	Charge air coolant pump
5	Left charge air radiator
6	Right charge air radiator
7	Radiator upper hose

Electrical power for the charge air coolant pump is supplied from the charge air coolant pump relay in the EJB (engine junction box) . When the relay is energized, it connects power from the battery to the charge air coolant pump. Operation of the relay is controlled by the ECM (engine control module) . The charge air coolant pump relay is energized continuously while the ignition is in power mode 6.

When the charge air coolant pump is running, coolant flows from the pump outlet through the charge air coolers, the charge air radiators and back to the pump intake.

CONTROL DIAGRAM



A = HARDWIRED.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2

3	Battery junction box
4	Auxiliary junction box
5	Engine junction box (charge air coolant pump relay)
6	Charge air coolant pump
7	Engine control module